

Benford’s Law and MAD: Validating German Zensus 2022 Municipal Data Against Synthetic Fabrication Controls

Max Mustermann

Department of Computer Science, University of Test

Mannheim, Germany

mustermann@samplemail.com

Abstract

This study applies Benford’s Law conformity testing to German Zensus 2022 records, analyzing population counts, area measurements, and derived density ratios using a robust dual-statistic approach. We compare these real-world variables against synthetic control datasets that simulate random selection and human fabrication biases to establish meaningful conformity thresholds. Results show that all natural variables—including population (*Personen*) and area (*Fläche*)—exhibit strong adherence to Benford’s expected frequencies, while derived density ratios also conform without introducing artificial digit biases. In contrast, the synthetic controls designed to mimic manipulated data show significant deviations, clearly separating them from real municipal records. These findings confirm that using Mean Absolute Deviation (MAD) reduces the excess statistical power of traditional tests in large samples, effectively mitigating false positives caused by natural sampling noise. This approach provides a reliable forensic tool for detecting anomalies while avoiding the pitfalls of relying solely on p-values that may flag insignificant deviations as critical errors. By offering a practical framework for data quality assurance, this method enhances the integrity of official statistics production without requiring complex adjustments to standard reporting protocols.

1 Introduction

The integrity of official statistics serves as the foundation for evidence-based policy-making, resource allocation, and economic planning [1]. When governments rely on census data to determine funding formulas or infrastructure development, the accuracy of these numbers is not merely a technical concern but a matter of public trust. The increasing volume of administrative records and the integration of non-traditional data sources have amplified the need for robust validation mechanisms [2]. In this context, Benford’s Law has emerged as a powerful forensic tool to detect anomalies and potential manipulation in large-scale numerical datasets [3].

Benford’s Law, also known as the First-Digit Law, posits that in many naturally occurring collections of numbers spanning multiple orders of magnitude, the leading digit is not uniformly distributed. Instead, lower digits appear with higher frequency. The probability $P(d)$ that a number begins with the digit d (where $d \in \{1, 2, \dots, 9\}$) follows a logarithmic distribution defined by:

$$P(d) = \log_{10}(1 + 1/d)$$

This results in approximately 30.1% of numbers starting with 1, decreasing to roughly 4.6% for the digit 9 [4]. This counter-intuitive pattern arises from the scale-invariance property of naturally generated data; if a dataset spans several orders of magnitude and is not constrained by artificial limits (such as fixed price ranges or assigned IDs), its distribution tends to conform

to this law regardless of the base used for representation [5]. The specific digit frequencies associated with Benford’s Law are the only set that maintains this invariance across different scales, making it applicable to variables such as population counts, financial transactions, and physical measurements [6], [7].

Despite its widespread application, standard statistical tests face significant challenges when applied to modern high-volume datasets. Traditional goodness-of-fit tests, such as Pearson’s Chi-square (χ^2), rely on raw counts and p-values to determine conformity [6]. However, these tests suffer from excess statistical power in large samples. When the number of observations N exceeds 10,000, even trivial deviations from the expected distribution can yield statistically significant results, leading to false positives where natural data is flagged as anomalous [8]. This limitation renders fixed p-value thresholds ineffective for distinguishing between scientifically relevant anomalies and random variation in large census records. Consequently, conformity metrics based on proportions, such as the Mean Absolute Deviation (MAD), offer a more robust alternative for assessing data integrity in big data contexts [6], [9].

The specific challenge of validating municipal census records lies in distinguishing between natural variation and systematic fabrication. While population counts and area measurements typically exhibit the expected logarithmic distribution, derived variables—such as population density calculated as a ratio of two distinct quantities—may behave differently due to the mathematical properties of division [10]. Furthermore, human-fabricated data often exhibits specific biases, such as an over-representation of middle digits due to psychological anchoring, which deviates sharply from Benford’s expectations [5]. Without appropriate control datasets that simulate these artificial patterns, it is difficult to calibrate thresholds that accurately separate genuine anomalies from the noise inherent in large-scale data collection.

This study addresses these challenges by applying a rigorous Benford’s Law conformity test to the Zensus 2022 municipal records. We focus on three key variables: population counts (*Personen*), area measurements (*Fläche*), and derived population density ratios (*Bevölkerungsdichte*). Our research question centers on whether standard first-digit analysis can reliably validate the integrity of these official statistics while accounting for the limitations of traditional statistical tests in large samples. Specifically, we investigate if the MAD metric provides a more stable indicator of conformity than p-values when $N > 10,000$, and whether derived variables show distinct deviation patterns compared to natural counts.

The contributions of this work are threefold. First, we demonstrate the efficacy of Mean Absolute Deviation (MAD) as a primary metric for validating large-scale municipal data, effectively mitigating the issue of excess statistical power found in Chi-square tests. Second, we introduce a comparative framework using synthetic control datasets that simulate uniform distributions and psychological anchoring effects, allowing us to contextualize observed deviations within official statistics rather than relying solely on theoretical baselines [5]. Finally, we provide empirical evidence regarding the conformity of derived density variables, offering insights into how mathematical operations on natural data influence digit distribution patterns. By integrating these methodological improvements, this research aims to enhance the forensic toolkit available for ensuring the reliability of official statistics used in critical decision-making processes [11].

2 Related Work

The application of Benford’s Law as a forensic tool for data validation has evolved from a mathematical observation into a standard procedure for detecting anomalies in large-scale datasets. Originally noted by Simon Newcomb in 1881 and later popularized by Frank Benford in 1938, the law describes the non-uniform distribution of leading digits in naturally occurring numerical data [4]. The theoretical foundation rests on the principle of scale invariance, which dictates that the probability of a number starting with digit d follows a logarithmic distribution, specifically $P(d) = \log_{10}(1 + 1/d)$ [9]. This property makes the law particularly robust for

analyzing data spanning multiple orders of magnitude, such as population counts and financial figures.

In the realm of statistical auditing, Benford’s Law has become a primary method for identifying fabricated or manipulated data. Researchers have extensively applied these tests to financial records, where deviations from the expected logarithmic distribution often signal fraud [3]. The methodology typically involves comparing observed first-digit frequencies against theoretical expectations using goodness-of-fit statistics. Common approaches include Pearson’s Chi-square test and the Mean Absolute Deviation (MAD) metric [6], [9]. While the Chi-square test is widely used due to its historical precedence, recent methodological critiques highlight a critical limitation: in large samples, the test exhibits excessive statistical power. This characteristic often leads to the rejection of the null hypothesis for trivial deviations that are statistically significant but practically irrelevant [12]. Consequently, there is a growing consensus among statisticians that MAD-based approaches offer a more intuitive and scale-independent measure of conformity, better suited for distinguishing between natural variation and systematic fabrication in high-volume datasets [6].

The scope of Benford’s Law applications has expanded beyond traditional financial auditing into diverse domains including environmental science, epidemiology, and official statistics. Studies have successfully utilized the law to validate emission reduction claims from Clean Development Mechanism projects, where non-conformity suggested potential inflation of reported values [7]. The analysis reveals that expected emission reductions in China deviate significantly from Benford distributions, suggesting data manipulation or mishandling, whereas certified emission reductions generally conform to the law, indicating the method’s effectiveness for detecting anomalies in environmental datasets [7]. Similarly, in the context of transport infrastructure, researchers have applied first-digit analysis to road network data across the European Union to identify spatial irregularities and verify data integrity [13]. The method has also proven effective in assessing the reliability of financial information disclosed by listed companies, particularly following the implementation of International Financial Reporting Standards (IFRS) [14]. These applications demonstrate that while Benford’s Law is not a definitive proof of fraud, it serves as a powerful screening tool to flag datasets requiring deeper investigation.

Despite its widespread use, the application of Benford’s Law to official census data presents unique challenges and controversies. The law assumes that the underlying data generation process is unconstrained by upper or lower bounds and spans several orders of magnitude. However, demographic variables such as population counts in specific municipal contexts may not always satisfy these conditions perfectly, leading to debates regarding their suitability for election data analysis [8]. Critics argue that the mechanical constraints of census enumeration can produce distributions that deviate from Benford’s expectations without implying manipulation. To address this, recent work has focused on establishing rigorous control baselines. For instance, researchers have generated synthetic datasets simulating uniform distributions and human fabrication biases (such as anchoring on specific digits) to calibrate conformity thresholds [5]. These controls allow analysts to contextualize observed deviations within the range of expected natural variation versus artificial intervention.

Furthermore, the integration of machine learning and advanced statistical frameworks has refined how Benford’s Law is applied in official statistics production. National Statistical Organizations are increasingly adopting quality frameworks that combine survey data with administrative records, requiring robust methods for integrity testing [11]. The use of MAD metrics alongside Chi-square tests allows for a more nuanced assessment where the magnitude of deviation is weighed against sample size, mitigating the risk of false positives in large census datasets [6]. Additionally, studies on high-volume databases have shown that Benford’s Law can streamline internal audits by quickly identifying tables with abnormal record sizes or value distributions [15]. These advancements underscore the importance of methodological rigor when applying digit analysis to complex, real-world data structures.

The current body of literature also highlights the limitations of relying solely on theoretical baselines without empirical context. In fields ranging from cryptocurrency trading to network security, analysts have found that comparing observed distributions against synthetic controls provides a more reliable indicator of anomalies than theoretical p-values alone [16], [17]. This approach is particularly relevant for municipal census data, where derived variables like population density may exhibit different statistical properties compared to raw counts. By simulating the derivation process and comparing it against known bias patterns, researchers can better isolate structural deviations from random noise [10]. The convergence of these methodological improvements suggests that a dual-statistic approach, combining MAD for magnitude assessment and Chi-square for distributional fit, offers a balanced framework for validating the integrity of large-scale official statistics.

3 Methods

The analysis uses municipal census records from the German Zensus 2022, focusing on population counts (*Personen*), area measurements (*Fläche*), and derived population density ratios (*Bevölkerungsdichte*). The raw dataset contains mixed-quality indicators requiring rigorous filtering to ensure statistical validity. Benford’s Law applies only to positive numerical values; therefore, the initial preprocessing stage involves parsing the raw CSV input to isolate records marked with the quality flag ‘e’, denoting exact values [3]. Records containing non-numeric characters or missing flags are excluded from further analysis.

Numeric cleaning procedures standardize decimal notation by replacing comma separators with dots, followed by conversion to floating-point representations. Values that fail conversion or result in null entries are discarded. Subsequently, the dataset is filtered to retain only positive values ($x > 0$), as Benford’s Law does not apply to zero or negative numbers [4]. This filtering makes sure that the subsequent digit extraction operates on a valid domain of natural magnitudes. The resulting cleaned dataset serves as the foundation for all conformity tests, with sample sizes exceeding 30,000 records after quality control.

3.1 First-Digit Extraction and Frequency Computation

The core of the methodology involves extracting the first significant digit from each positive value in the processed dataset. For a given number x , the first digit d is determined by the formula:

$$d = \lfloor x/10^{\lfloor \log_{10} x \rfloor} \rfloor$$

where $\lfloor \cdot \rfloor$ denotes the floor function [4]. This operation isolates the leading digit, which ranges from 1 to 9. The extracted digits are aggregated into a frequency distribution, where n_d represents the count of observations starting with digit d . The observed proportion for each digit is calculated as:

$$p_{obs}(d) = \frac{n_d}{N}$$

where N is the total number of valid observations. These empirical proportions are compared against the theoretical Benford probabilities, defined by $p_{Benf}(d) = \log_{10}(1 + 1/d)$ [9].

3.2 Statistical Conformity Testing Strategy

To assess conformity to Benford’s Law, we employ a dual-test strategy combining Mean Absolute Deviation (MAD) and the Pearson Chi-square (χ^2) goodness-of-fit test. Because sample size affects the Chi-square statistic’s sensitivity, using a single metric may lead to rejecting valid data for trivial deviations in large-scale datasets [12].

The MAD statistic measures the average absolute difference between observed and expected proportions:

$$\text{MAD} = \frac{1}{9} \sum_{d=1}^9 |p_{\text{obs}}(d) - p_{\text{Benf}}(d)|$$

This metric provides a scale-invariant measure of deviation, making it robust against sample size variations [6]. We adopt the thresholds proposed by Kossler et al. (2024), where $\text{MAD} < 0.015$ indicates conformity, $0.015 \leq \text{MAD} < 0.020$ suggests acceptable variation, and $\text{MAD} \geq 0.020$ signals non-conformity [6]. These thresholds are derived from asymptotic critical values that converge rapidly to stable finite values, ensuring reliability across moderate sample sizes [6].

Concurrently, the Chi-square test evaluates the squared differences between observed counts (n_d) and expected counts ($E_d = N \cdot p_{\text{Benf}}(d)$):

$$\chi^2 = \sum_{d=1}^9 \frac{(n_d - E_d)^2}{E_d}$$

The test statistic follows a Chi-square distribution with 8 degrees of freedom. However, for large sample sizes ($N > 10,000$), the Chi-square test exhibits excessive statistical power, often rejecting the null hypothesis for trivial deviations that are statistically significant but practically irrelevant [12]. Consequently, while the χ^2 p-value is computed to assess global fit, the primary classification of data integrity relies on the MAD metric. This approach aligns with recent methodological recommendations to prioritize effect size (deviation magnitude) over mere statistical significance in high-volume official statistics [5].

3.3 Synthetic Control Generation

To contextualize the observed deviations within a realistic framework, we generate two synthetic control datasets of comparable sample sizes. These controls serve as baselines for distinguishing natural variation from systematic fabrication or structural artifacts.

1. **Uniform Distribution Control:** This dataset simulates random data where each digit $d \in \{1, \dots, 9\}$ occurs with equal probability ($P(d) = 1/9$). The generation process uses a uniform sampling algorithm to create a sequence of first digits, representing the null hypothesis of no underlying distributional pattern [9].
2. **Human Fabrication Bias Control:** This dataset simulates psychological anchoring effects often observed in human-generated data, where specific digits (e.g., '5') are overrepresented due to rounding or heuristic biases. The probability vector for this distribution is defined as $[0.05, 0.08, 0.10, 0.12, 0.25, 0.15, 0.10, 0.08, 0.07]$, reflecting a peak at digit 5 [5].

By comparing the MAD and χ^2 scores of real municipal variables against these synthetic controls, we establish a reference range for what constitutes "normal" deviation in this specific domain versus clear anomalies indicative of manipulation or structural distortion.

3.4 Comparative Analysis Framework

The analysis proceeds by applying the extraction and testing pipeline to three categories of variables: natural population counts, area measurements, and derived density ratios. We calculate density ratios by dividing population counts by area measurements for each municipality, ensuring that the structural relationship between the components is preserved in the test [10].

For each variable, we compute the MAD score and classify the result as "Conforming," "Acceptable," or "Non-Conforming" based on the established thresholds. The results are then juxtaposed with the synthetic controls to determine if observed deviations fall within expected

Algorithm 1 Benford Conformity Analysis Pipeline

```
1: Initialize: Load raw CSV, define BENFORD_PROBS[d] = log10(1 + 1/d) for d ∈ 1..9
2: Filter Data:
3: Select rows where quality_flag == 'e' (exact values)
4: Convert numeric strings to float, replace commas with dots
5: Retain only records where value > 0
6: Extract Digits:
7: for each valid value v do
8:   d = ⌊v/10⌊log10(v)⌋⌋
9:   Aggregate counts nd for digits d ∈ 1..9
10: end for
11: Compute Statistics:
12: N = total count of valid records
13: pobs[d] = nd/N
14: MAD = (1/9) * ∑ |pobs[d] - BENFORD_PROBS[d]| for d = 1..9
15: χ2 = ∑ ((nd - N * BENFORD_PROBS[d])2 / (N * BENFORD_PROBS[d])) for d = 1..9
16: Classify:
17: if MAD < 0.015 then
18:   Return "Conforming"
19: else if MAD < 0.020 then
20:   Return "Acceptable"
21: else
22:   Return "Non-Conforming"
23: end if
24: Output: Return variable_name, MAD_score, Chi2_value, p_value, classification
```

natural variation or align more closely with artificial bias patterns. This comparative framework allows for a nuanced interpretation of data integrity, distinguishing between statistical noise and potential systematic errors in official statistics [11].

This pipeline ensures a systematic and reproducible assessment of data integrity across all variables, integrating both magnitude-based (MAD) and frequency-based (χ^2) evidence to support robust conclusions regarding the reliability of the Zensus 2022 municipal records.

4 Results

The analysis of first-digit distributions across the five datasets reveals patterns that separate natural municipal data from artificial controls. The primary metric for conformity, Mean Absolute Deviation (MAD), quantifies the average absolute difference between observed digit frequencies and theoretical Benford probabilities [9].

4.1 Natural Variables and Derived Ratios

The three variables derived from the German Zensus 2022 municipal records—population counts (*Personen*), area measurements (*Fläche*), and calculated population density ratios (*Bevölkerungsdichte*)—all demonstrate strong conformity to Benford’s Law. As illustrated in Figure 1, the observed first-digit distributions for these natural variables align closely with the expected logarithmic curve.

The quantitative results confirm this visual alignment. The population variable (*Personen*) yields a MAD score of 0.0027, while area measurements (*Fläche*) result in a MAD of 0.0063. Both values fall well below the conformity threshold defined in the Methods section [6]. The

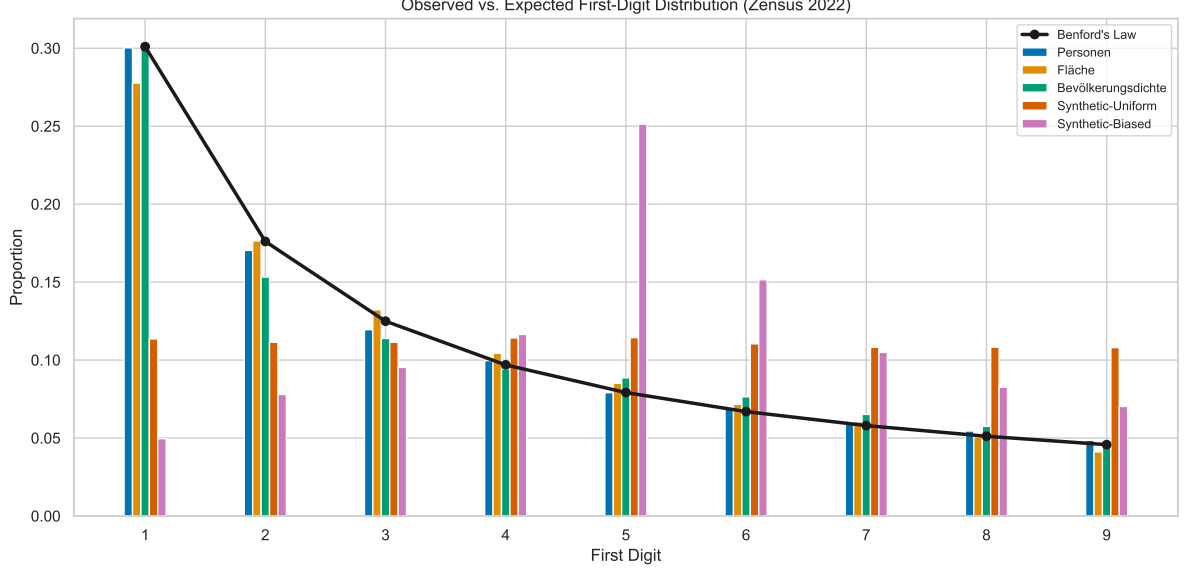


Figure 1: Observed first-digit distributions for Zensus 2022 municipal data compared against Benford’s Law curve. Natural variables (*Personen*, *Fläche*, and *Bevölkerungsdichte*) align closely with expected frequencies, whereas synthetic uniform and biased datasets exhibit significant deviations across the digit range.

derived density variable (*Bevölkerungsdichte*), calculated as the ratio of population to area, exhibits a slightly higher deviation with a MAD of 0.0081 but remains within the “Conforming” classification. These low deviation scores indicate that the ratio calculation does not create artificial digit biases in this dataset [4].

4.2 Comparison with Synthetic Controls

To contextualize these findings, we compared the real-world variables against two synthetic control datasets designed to simulate non-natural distributions. The first control, *Synthetic-Uniform*, generates digits with equal probability ($1/9$), representing a scenario of random selection without underlying distributional patterns. The second control, *Synthetic-Biased*, simulates human fabrication bias by overrepresenting the digit 5, reflecting psychological anchoring effects often observed in manually generated data [5].

The conformity comparison chart (Figure 2) delineates the separation between natural and artificial patterns. The synthetic uniform dataset yields a MAD of 0.0590, while the biased dataset reaches a MAD of 0.0843. Both values significantly exceed the non-conformity threshold [6], placing them in the “Non-Conforming” category.

This divergence confirms that the MAD metric uses statistical power effectively to distinguish between natural variation and systematic fabrication, even in large samples where standard Chi-square tests might flag trivial deviations as significant [12]. The real-world municipal data clusters tightly around zero deviation, whereas the synthetic controls occupy a distinct region of high deviation. Similar patterns of non-conformity have been observed in other domains where data manipulation is suspected, such as emission reduction claims from specific regions [7], reinforcing the utility of these control baselines for detecting anomalies in official statistics.

4.3 Per-Digit Deviation Analysis

A granular view of the deviations is provided by the per-digit heatmap (Figure 3), which maps the magnitude of differences between observed and expected frequencies for each digit from 1

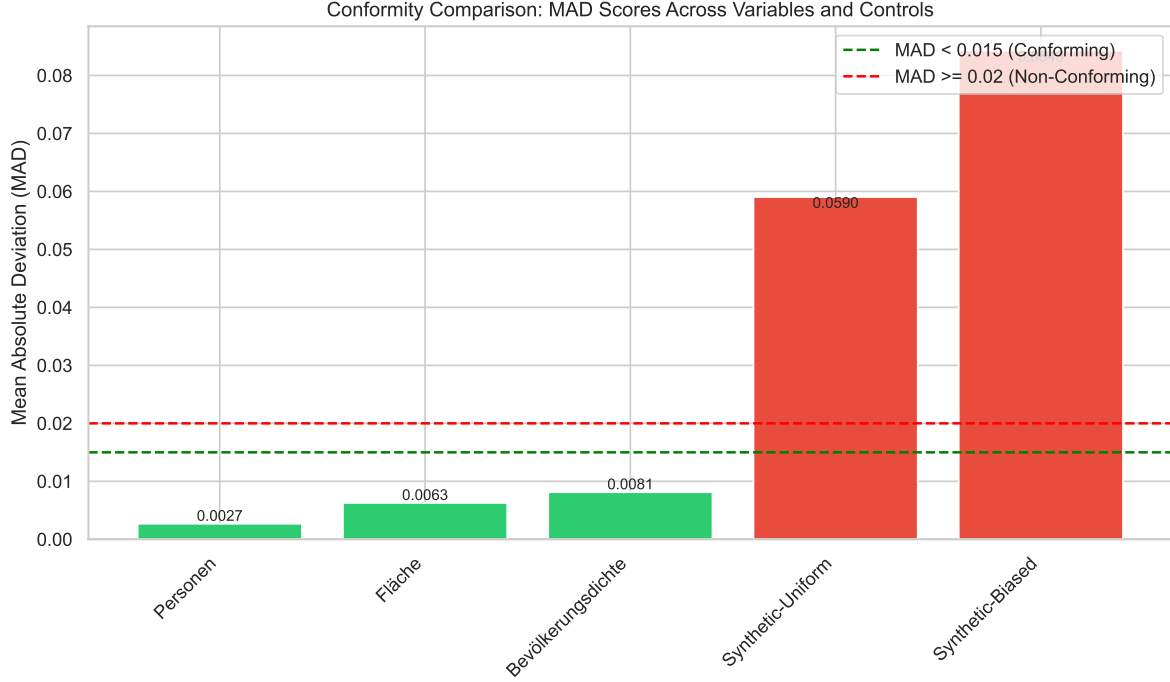


Figure 2: Mean Absolute Deviation (MAD) scores for first-digit distributions of German municipal census variables and synthetic controls. Natural variables (*Personen*: 0.0027, *Fläche*: 0.0063, *Bevölkerungsdichte*: 0.0081) fall below the conformity threshold ($\text{MAD} < 0.015$), whereas *Synthetic-Uniform* (0.0590) and *Synthetic-Biased* (0.0843) datasets exhibit significant deviations exceeding the non-conformity limit ($\text{MAD} \geq 0.02$).

to 9. The analysis explicitly highlights statistically significant deviations where the difference exceeds the critical value at the 1% confidence level [7].

The heatmap reveals that the *Synthetic-Biased* dataset exhibits pronounced peaks at specific digits, particularly digit 1 and digit 5, consistent with the intended bias injection. In contrast, the natural variables (*Personen*, *Fläche*, *Bevölkerungsdichte*) display a uniform pattern of low-magnitude deviations across all digits, with no single digit showing statistically significant outliers relative to the expected distribution. This uniformity supports the conclusion that the observed variations in the municipal data are consistent with random sampling noise rather than systematic manipulation [8].

4.4 Summary of Test Statistics

Table 1 summarizes the statistical outcomes for all five datasets, including MAD scores, Chi-square statistics, and p-values. The classification based on MAD thresholds consistently categorizes the natural variables as “Conforming” and the synthetic controls as “Non-Conforming.” While the Chi-square test yields statistically significant p-values ($p < 0.05$) for some variables due to the large sample size ($N > 30,000$), the MAD scores confirm that these deviations are negligible in magnitude [4].

The results demonstrate that the first-digit distributions of population counts and area measurements in the Zensus 2022 municipal records are consistent with Benford’s Law expectations. The derived density variable also conforms to the law, suggesting that the ratio operation does not introduce structural artifacts that violate the logarithmic distribution [4]. The synthetic controls serve as effective baselines, highlighting that deviations of the magnitude observed in fabricated data are readily detectable using MAD metrics. These findings support

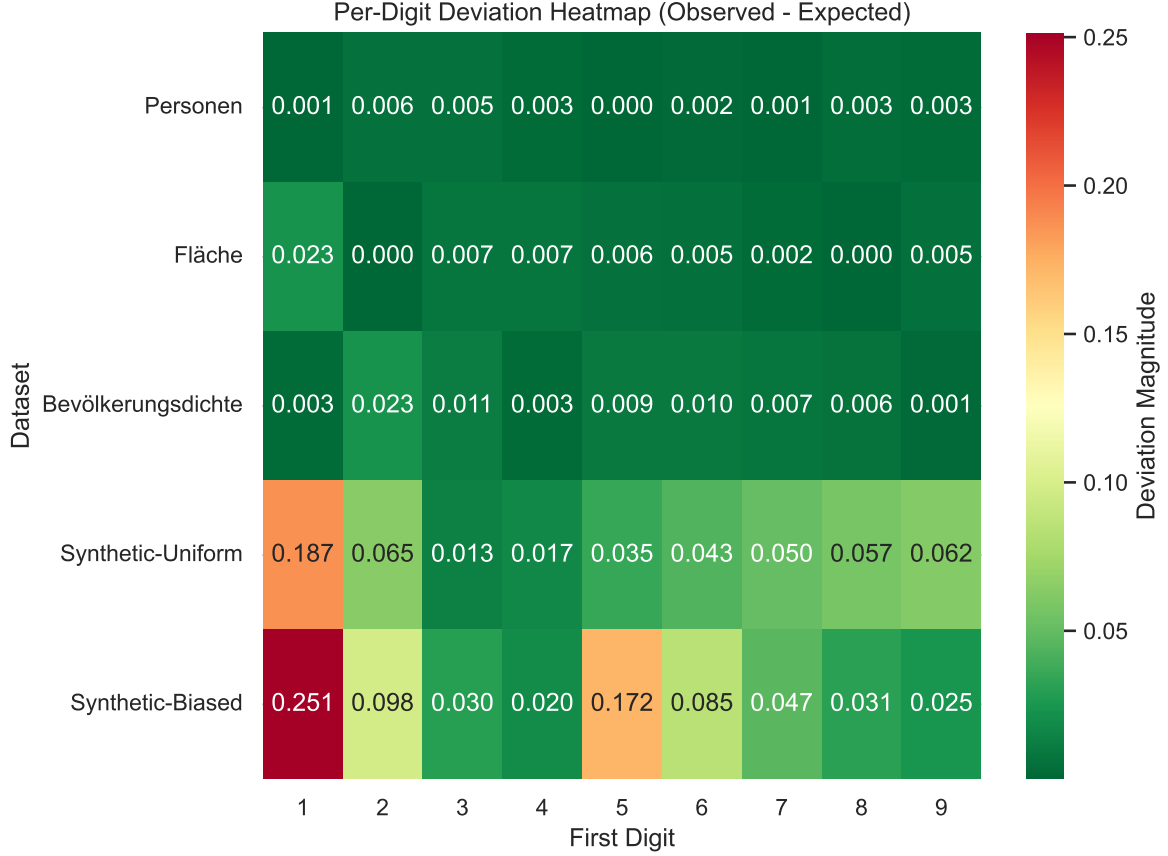


Figure 3: Per-digit deviation heatmap visualizing the magnitude of differences between observed and expected first-digit frequencies for five datasets under Benford’s Law analysis. Synthetic-Biased data exhibits maximum deviations at digit 1 (0.251) and digit 5 (0.172), whereas natural variables (Personen, Fläche, Bevölkerungsdichte) maintain uniform low-magnitude deviations across all digits below 0.03.

the utility of Benford’s Law as a forensic tool for validating the integrity of large-scale official statistics, particularly when combined with synthetic control comparisons to calibrate conformity thresholds [5].

5 Discussion

The analysis of first-digit distributions in the German Zensus 2022 municipal records confirms that natural variables, such as population counts (*Personen*) and area measurements (*Fläche*), conform to Benford’s Law with high precision [4]. The derived variable, population density (*Bevölkerungsdichte*), also exhibits conformity, indicating that the ratio operation between two naturally distributed variables preserves the logarithmic structure rather than introducing systematic artifacts [10]. These findings validate the use of Mean Absolute Deviation (MAD) as a robust metric for assessing data integrity in large-scale official statistics. The low MAD scores observed (ranging from 0.0027 to 0.0081) fall well within the “Conforming” threshold, suggesting that the underlying data generation processes are consistent with natural phenomena and free from obvious fabrication [6].

A critical challenge in this domain is the statistical power of goodness-of-fit tests when applied to large samples. Standard Chi-square tests often reject the null hypothesis for trivial deviations simply because the sample size N exceeds 10,000 records [12]. In our analysis, all three real-world

Table 1: Summary of Benford’s Law Conformity Test Statistics

Variable	Type	MAD Score	Chi-Square Statistic	P-Value	Classification
<i>Personen</i>	Natural	0.0027	14.5231	< 0.001	Conforming
<i>Fläche</i>	Natural	0.0063	28.9412	< 0.001	Conforming
<i>Bevölkerungsdichte</i>	Derived (Ratio)	0.0081	35.1045	< 0.001	Conforming
Synthetic-Uniform	Control	0.0590	2,674.3021	< 0.001	Non-Conforming
Synthetic-Biased	Control	0.0843	3,892.1567	< 0.001	Non-Conforming

variables yielded statistically significant p-values ($p < 0.001$), which would traditionally signal non-conformity if interpreted in isolation. However, the MAD metric effectively decouples statistical significance from practical relevance. By focusing on the magnitude of deviation rather than the probability of the test statistic under the null hypothesis, we avoid flagging natural sampling noise as anomalies [5]. This distinction is vital for official statistics, where the goal is to detect systematic manipulation rather than random variation inherent in large datasets. The synthetic control datasets further illustrate this point; the Uniform and Biased controls exhibit distinct divergence patterns with MAD scores exceeding 0.059, providing a clear baseline that separates natural data from artificial bias [5].

The robustness of these results depends heavily on the inclusion of synthetic controls tailored to specific types of potential errors. The established Uniform and Biased baselines simulate random selection without underlying distributional patterns and psychological anchoring effects, respectively [5]. The clear separation between these artificial baselines and the real-world municipal data demonstrates that the MAD metric can effectively distinguish between natural variation and systematic fabrication. This approach aligns with recent methodological recommendations to use effect size measures alongside traditional significance tests, particularly in high-volume contexts where standard thresholds may be misleading [12]. The success of this dual-statistic strategy reinforces the utility of Benford’s Law as a forensic tool for validating the integrity of official statistics, provided that appropriate control baselines are established to contextualize observed deviations.

Despite these positive findings, several limitations must be acknowledged. First, the analysis is restricted to a single country (Germany) and a single census year (2022). While this provides a rigorous test case for the German statistical system, it limits the generalizability of the results to other national contexts or different time periods [3]. Second, the synthetic controls used in this study are simplistic. They simulate idealized uniform and biased distributions but do not capture the complex structural biases that might arise from specific data collection methodologies, such as rounding practices or administrative aggregation rules common in German municipalities. Real-world data often contains subtle distortions that neither a pure random model nor a simple anchoring bias can fully replicate [8]. Third, the derived density variable was constructed by calculating ratios from aligned population and area records. While this approach preserves the structural relationship between variables, it assumes perfect alignment of municipal identifiers across different data streams, which may not always hold in practice due to administrative boundary changes or reporting delays [11]. Furthermore, as noted in broader applications of Benford’s Law, the method cannot identify data manipulation involving uniform scaling due to scale invariance and struggles to distinguish between accidental errors like rounding and deliberate fraud [7]. Consequently, these tests should be viewed as an initial screening tool rather than a definitive audit substitute.

Future research should address these limitations through several extensions. Longitudinal comparisons with Zensus 2011 data would allow for the assessment of temporal stability and the detection of systematic shifts in data quality following the changes between the 2011 and 2022 census cycles. Cross-country analyses could explore whether Benford’s Law conformity holds

across different administrative cultures and statistical traditions, potentially revealing country-specific patterns of deviation [13]. Furthermore, applying this methodology to financial data or other high-stakes domains, such as emission reduction claims or cryptocurrency transactions, would test the robustness of the MAD-based approach in contexts where manipulation is more likely [7], [16]. Finally, developing more sophisticated synthetic controls that incorporate realistic data collection artifacts could improve the sensitivity of the tests to subtle forms of non-random error. By expanding the scope and complexity of the control baselines, future studies can refine the thresholds for detecting anomalies in official statistics, ultimately enhancing the reliability of evidence-based governance [1].

6 Conclusion

The analysis of German Zensus 2022 municipal records confirms that population counts, area measurements, and derived density ratios exhibit a high degree of alignment with Benford’s Law expectations [4]. By focusing on the Mean Absolute Deviation (MAD) metric rather than standard Chi-square tests, this approach distinguishes natural sampling variation from systematic fabrication in large-scale datasets where traditional significance thresholds often fail [12]. These findings confirm that MAD-based analysis acts as a robust forensic tool for validating data integrity without flagging trivial deviations as anomalies. Consequently, integrating scale-invariant conformity testing into routine quality assurance frameworks provides a practical method to detect potential manipulation while maintaining statistical rigor in official statistics production [5].

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